

PIA 412 Course Outline

PIA 412 – The AICTEM PIAS System

The whole chain, end to end, on one machine.

School	04 – Public Investment
Hours	40 (indicative band 36-45)
Version	1.0 (2026-09-05)
Model	AICTEM PIAS, D:/EDIP/models/PIASModelling
Reference country	Utopia (UTP), UTD 2024 prices
Prerequisites	All eleven preceding courses. PIA 412 assumes you already know what an ENPV is, what a switching value is, what an expected annual loss is and what a solver status licenses; it is about the system that computes them and the record it keeps of having done so.
Audience	Analysts who will operate PIAS, the officials who will rely on its outputs, and the reviewers who have to decide whether an output may be quoted
Pass mark	60 %

What this course is

Eleven courses have taught what the numbers mean. This one is about the machine: 31,469 lines across 84 files and 13 subsystems, 25 engines, 90 database tables, and 577 assertions in 82 testsets that say what has been checked.

The central discipline of the course is that a system's status words are not its findings. Every one of the 11 macro-fiscal equilibrium loops in this register terminated at iteration 1 and 10 of them are recorded in state CONVERGED. Read the status column and the portfolios have been reconciled with the fiscal frame. Read the note attached to the same row and they have not: the loop was never requested, no debt path was supplied on any iteration, and no ceiling was ever adjusted. The system says so plainly, in a field most readers never open.

The second discipline is that a schema is a claim and a row count is a fact. PIAS declares 90 tables and holds rows in 54 of them; 36 are empty, and this course classifies every one of the 36 by the reason it is empty. There are 6 distinct reasons in this register: an engine that was not run, an input never captured, an asset not yet built, an interface no person has used, a system never connected, and a case this register does not contain. Those are six different sentences, and a commissioning report that writes one of them for all 36 is wrong.

Not one of the 11 sample projects is recommended to proceed. 5 are rejected, 5 require additional study and 1 is returned for revision, and most of the reasons given are absences of evidence rather than failures of economics. A system that recommends nothing when nothing has been evidenced is working correctly, and explaining why is the last thing this school teaches.

What a graduate can do

Describe the PIAS chain from a project record to a recommendation and name what each step consumes and writes; read a schema against its row counts and classify an empty table by the reason it is empty; audit the provenance carried by a result row and say what it does and does not pin down; reconcile a recommendation against the gate assessments that produced it; read an equilibrium loop's state alongside its note and write the sentence the pair supports; say what an integration that never ran leaves untested; audit an event trail for attribution and reason; read a test suite and state what it establishes and what it cannot; reproduce a published result from the register on one machine; and write a commissioning report that distinguishes what the system does from what this instance has done.

What this course does not claim

This course does not teach Julia, SQL or software engineering. It reads a system rather than building one, and every exercise is performed against exported tables with the standard library.

It does not audit the correctness of the appraisal arithmetic. PIA 403 to PIA 411 did that, engine by engine. Here the question is whether the system's own record of what it computed can be relied on, which is a different question and is usually the one nobody asks.

It does not certify the instance. A commissioning report is an output of this course and it is a statement of findings, not an accreditation. The distinction matters because 36 of the 90 tables are empty and a report that reads as approval will be quoted as one.

It does not supply the macro-fiscal frame. The equilibrium loop expects a fiscal path from FINEX; none was published for the scenario in this register, and 0 of the 11 loops carry a debt path. The course teaches how to read that absence, not how to fill it.

Learning outcomes

1. Describe the PIAS chain from a project record to a recommendation.
2. Read a schema against its row counts.
3. Classify an empty table by the reason it is empty.
4. Audit the provenance carried by a result row.
5. Reconcile a recommendation against its gate assessments.
6. Say what a gate status of PENDING licenses and what it forbids.
7. Read an equilibrium loop's state alongside its note.
8. Say what an integration that never ran leaves untested.
9. Audit an event trail for attribution and reason.
10. State what a test suite establishes and what it cannot.
11. Reproduce a published result from the register on one machine.
12. Write a commissioning report that separates a defect from a gap.

Module structure

Module	Title	Sessions	Hours	Laboratory	Manual	Assessment
M01	The system, measured	1	1.75	–	1	Quiz M01
M02	The schema and the counts	1	4.25	Lab 01	2	Quiz M02
M03	The chain, step by step	1	1.75	–	3	Quiz M03
M04	Provenance on every row	1	4.25	Lab 02	4	Quiz M04
M05	The gates the system records	1	1.75	–	5	Quiz M05
M06	The recommendation	1	4.25	Lab 03	6	Quiz M06
M07	The loop that converged	1	4.25	Lab 04	7	Quiz M07
M08	Integrations that never ran	1	1.75	–	8	Quiz M08
M09	The audit trail	1	1.75	–	9	Quiz M09
M10	What the tests establish	1	4.25	Lab 05	10	Quiz M10
M11	Reproducing a result	1	4.25	Lab 06	11	Quiz M11
M12	The commissioning report	1	1.75	–	12	Quiz M12

Laboratories

Lab	Title	Hours	Deliverable
1	Inventory the schema against its counts, and classify every empty table	2.5	The inventory
2	Audit the three provenance layers, and write the block for one published number	2.5	The provenance audit
3	Reconcile every recommendation with the gate grid, and find the failure that vanishes	2.5	The gate audit
4	Read the equilibrium loop’s state alongside its note, and audit the trail	2.5	The loop and trail audit
5	Read the test suite, and write what it establishes and what it cannot	2.5	The test audit
6	Reproduce three quantities from their own inputs, and classify every difference	2.5	The reproduction report

Assessment

Component	Weight
Knowledge quizzes	15 %
Laboratory exercises	30 %
Applied assignments	25 %
Capstone / final project	30 %

Applied assignments

ID	Title	Weight	Deliverable
A1	The inventory and the empty tables	25	Marked on the classification and on the sentences it supports
A2	The provenance audit	25	Marked on the gap-against-defect distinction
A3	Gates, recommendations and the loop	25	Marked on quoting the state and the note together
A4	The commissioning report	25	Marked on the three kinds of finding and on the certification limit

Final project

A commissioning report on this instance of PIAS. It must contain: an inventory of the schema against its row counts, with every empty table classified by the reason it is empty; a provenance audit stating what share of result rows carry an engine version, a parameter set version and a dataset version, and what each of those pins down; a trace of one project through every step of the chain, table by table, with row counts; a reconciliation of that project's recommendation against its gate assessments; a reading of the equilibrium loop that quotes the state and the note together and writes the sentence the pair supports; a statement of what the test suite establishes and what it does not; and a closing page of findings, each marked as a defect, a gap in the data, or a capability that is declared and unused – which are three different things and are routinely written as one.

Folder contents

Every course in the Academy carries the same section 5 folder. See `README.md` for what each item holds and `COURSE_BUILD_REPORT.md` for what was measured.